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Machine Learning Assignment 4.

EM Algorithm parameter Estimation:- EM Algo is used when we want to maximise $P(X|\theta)$ but there are hidden variable such as Z so instead of directly solving $\log P(x|\theta)$, EM works with $\log P(X, Z|\theta)$.

it Alternates b/w

E-Step :- estimate hidden structure.

M-step :- Optimize Parameters

Use cases → Missing data.

Missing Cluster labels etc.

given model in problem:- Two coins C_0, C_1

Z hidden variable : which coin was used in each trial.

observed data outcomes $D = \{HHH, HHT, TTT\}$

each trial :

→ Pick coin C_0 with probability K else C_1

→ toss 3 times.

given parameters. $K = P(C_0)$

$P_0 = P(H|C_0)$

$P_1 = P(H|C_1)$.

we don't know which coin generated outcome Hence hidden variable:

$$z_i = \begin{cases} 1 & \text{trial } i \text{ used } C_0 \\ 0 & \text{trial } i \text{ used } C_1 \end{cases}$$

maximise the likelihood $\log P(D|\theta)$

EM maximise the likelihood by E Step and M step.

if if we knew which coin was used :

$$P(D, Z|\theta) = \prod_i P(z_i)\, P(x_i | z_i)$$

example

$$P(x_i|C_0) = P_0^H \cdot (1-P_0)^T$$

$$P(x_i|C_1) = P_1^H (1-P_1)^T$$

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E step Compute responsibilities

$$\gamma_i = P(C_0 | x_i)$$

Using Bayes theorem

$$\gamma_i = \frac{K \cdot P(x_i | C_0)}{K P(x_i | C_0) + (1-K) P(x_i | C_1)}$$

M Step : update parameters:-

$$K_{new} = \frac{1}{N} \sum_j \gamma_i$$

$$P_{0new} = \frac{\sum_j \gamma_i (H_i)}{\sum_j \gamma_i \cdot 3}$$

$$P_{1new} = \frac{\sum_j (1-\gamma_i)(H_i)}{\sum_j (1-\gamma_i) \cdot 3}$$

<u>Computing the likelihood</u>

1) HHH case : $P(HHH | C_0)$: $(0.6)^3 = 0.216$

$P(HHH | C_1)$: $(0.1)^3 = 0.001$

2) HHT case : $P(HHT | C_0)$: $(0.6)^2 (0.4) = 0.144$

$P(HHT | C_1)$ $(0.1)^2 (0.9) = 0.009$

3) TTT case : $P(TTT | C_0)$: $(0.4)^3 = 0.064$

$P(TTT | C_1)$: $(0.9)^3 = 0.729$

Computing γ_i

$$\gamma_1 = \frac{0.5 \times 0.216}{0.5 \times 0.216 + 0.5 \times 0.001} = \frac{0.108}{0.1085} \approx 0.9954$$

$$\gamma_2 = \frac{0.5 \times 0.144}{0.5 \times 0.144 + 0.5 \times 0.009} \approx 0.9412$$

$$\gamma_3 = \frac{0.5 \times 0.064}{0.5 \times 0.064 + 0.5 \times 0.729} = \frac{0.032}{0.3965} \approx 0.0807$$

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M step update Params:-

$$K_{new} = \frac{0.9954 + 0.9412 + 0.0807}{3} = 0.6724$$

weighted heads (H)

$$P_0 = \frac{(0.9954 \times 3 + 0.9412 \times 2 + 0.0807 \times 0)}{}$$

$$\Rightarrow 4.8686.$$

Total tosses $= 3(0.9954 + 0.9412 + 0.0807) = 6.052$

$$P_{0\,new} = \frac{4.86}{6.052} = 0.8046$$

Update P_1 : $1-Y = (0.0046, 0.0588, 0.1175) = 0.314$

total tosses :

$$= 3(0.0046 + 0.0588 + 0.9193) = 2.947$$

$$P_{1\,new} = \frac{0.314}{2.947} \approx 0.0446.$$

after 1 EM iteration.

$K_{new} = 0.6724$

$P_{0\,new} = 0.8046$

$P_{1\,new} = 0.0446$

if $P_0 \uparrow$ more heads

$P_1 \uparrow$ more tails

$K \uparrow$ more data assignment to C_0.